LAB 02

i. Identify **actors** and the **system boundary** based on the Lab 01 scenario.

Actors:

- Borrower (Teacher/Student): Searches for equipment, sends borrowing requests, checks request progress, and returns borrowed items.
- IT Staff: Reviews and approves requests, updates equipment status, and confirms returned items.
- Administrator: Manages users, updates system data, and creates reports.
- Notification Service: Sends system or email alerts to users.
- System Timer: Automatically runs scheduled tasks like sending reminders or checking for overdue items.

System Boundary:

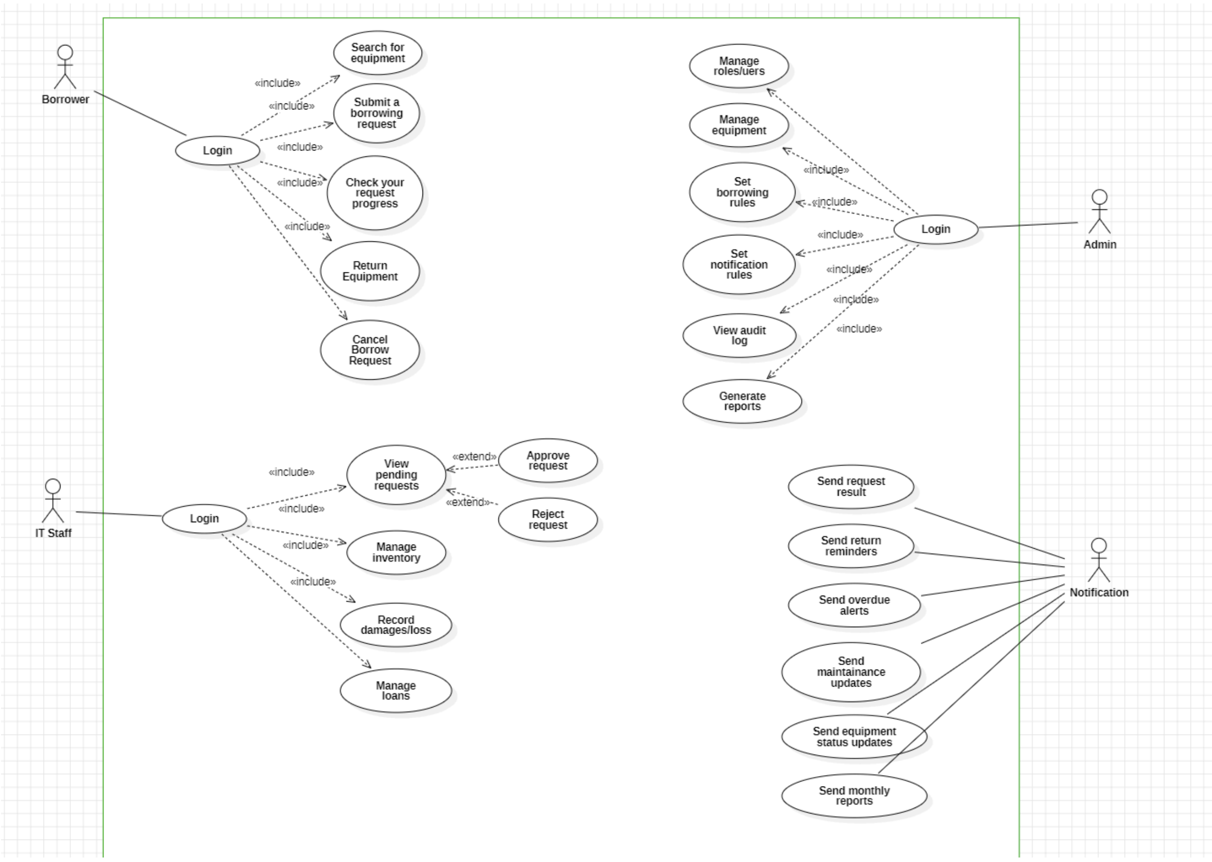
- Online submission of borrowing requests
- Approval and scheduling workflow
- Real-time tracking of equipment status
- Automated due date reminders and notifications
- Reporting and analytics tools
- Role-based user access and management

ii. List and group **business use cases** according to each actor's goals.

Actors	Use Cases
Borrower (Teacher/Student)	• Login/Logout • Find equipment • Submit a borrowing request • Check your request progress • View your past borrowings • Ask to extend your borrowing time • Cancel a request that's still pending • Report problems with borrowed items • Return equipment
IT Staff	• Login/Logout • See all borrowing requests • Approve request • Reject/Reschedule request • Manage equipment inventory • Track all equipment status • Record damaged or lost items • View loans process

	• Update loans process
Administrator	• Login/Logout • Manage roles/users • Manage equipment • Set borrowing rules • Set notification preferences • View activity log • View reports/analytics
Notification Service	• Send request result (approved/rejected) • Send return reminders • Send overdue alerts • Send maintenance updates • Send equipment status updates • Send monthly reports to Admin or IT staff
System Timer	• Automatically sends reminder emails and overdue alerts

iii. Model a **Use Case Diagram** (including *include/extend* relationships when appropriate).



iv. Write **fully dressed use case specifications** for 2–3 key use cases (*Borrow Equipment, Approve Request, Return Equipment*).

Approve Equipment

Field	Example Content
Use Case ID / Name	Approve Request
Scope / System	Equipment Borrowing Management System
Level	User Goal
Primary Actor	IT Staff
Supporting Actors	Notification Service
Stakeholders & Interests	IT Staff – wants clear and complete request details; Borrower – wants a fast decision; Admin – monitors system activity.
Goal in Context	To review, approve, reject borrowing requests.
Preconditions	There are pending borrowing requests in the system.
Trigger	IT Staff opens a pending request for review.

Main Success Scenario (Basic Flow)	1. System lists pending requests. 2. IT Staff reviews the borrower's details and purpose. 3. System checks item availability. 4. IT Staff selects "Approve," "Reject," or "Reschedule." 5. System updates the request status. 6. Borrower receives a notification about the result.
Extensions (Alternate Flows / Exceptions)	3a. Item not available → IT Staff suggests another time slot. 4a. Missing information → System prevents submission until complete.
Postconditions / Guarantees	Request status updated and borrower notified.
Special Requirements (NFR)	Changes visible immediately; system logs the decision.
Technology / Data Variations	Approval page may include a calendar view or equipment list.
Assumptions	IT Staff has appropriate system access.
Notes / Issues	Future enhancement: automatic approval for standard requests.

Returning Equipment

Field	Example Content

Use Case ID / Name	Return Equipment
Scope / System	Equipment Borrowing Management System
Level	User Goal
Primary Actor	Borrower
Supporting Actors	IT Staff, Notification Service, System Timer
Stakeholders & Interests	Borrower – wants quick return confirmation; IT Staff – ensures items are complete and functional; Admin – tracks returned items and conditions.
Goal in Context	To record the return of borrowed equipment and update its availability.
Preconditions	Equipment is currently borrowed and has not yet been marked as returned.
Trigger	Borrower submits or hands in the equipment for return.
Main Success Scenario (Basic Flow)	1. Borrower returns the equipment to IT Staff. 2. IT Staff inspects the item's condition. 3. The system updates its status to "Returned." 4. System Timer stops any overdue tracking. 5. Borrower receives a confirmation message.

Extensions (Alternate Flows / Exceptions)	2a. Item damaged or incomplete → IT Staff records issue and updates system. 3a. Returned late → System marks as “Overdue” and records penalty if needed.
Postconditions / Guarantees	Equipment marked as “Available”; record updated with return date.
Special Requirements (NFR)	Return confirmation appears within 5 seconds; actions logged for auditing.
Technology / Data Variations	Barcode or QR code scan may be used to record returns faster.
Assumptions	IT Staff is available and system is online during return.
Notes / Issues	Future improvement: generate automatic digital return receipt.

v. Ensure that the model is consistent with system goals (online request, approval workflow, real-time tracking, reminders, reporting).